Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

There are p letters a, q letters b, r letters c. The number of ways of selecting k letters out of these if $p < k < q < r$ is

(A) $\frac{1}{2}(p+1)^2 - k$

(B) $\frac{1}{2}(p+1)(2k-p)$

(C) $\frac{1}{3}(p+1)(q+1)(r+1) - k$

(D) none of these

Q.2

The number of integers x, y, z, w such that $x+y+z+w = 20$ and $x, y, z, w \geq -1$, is

(A) ${}^{24}C_3$

(B) ${}^{25}C_3$

(C) ${}^{26}C_3$

(D) ${}^{27}C_3$

Q.3

The number of ways of putting 5 identical balls in 10 identical boxes, if not more than one can go in a box is

(A) ${}^{10}P_5$

(B) ${}^{10}C_5$

(C) 10^5

(D) 1

Q.4

The number of ways distributing 8 identical balls in 3 distinct boxes so that none of the boxes is empty is

(A) 3^8

(B) 21

(C) 5

(D) ${}^{8}C_3$

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Q.5

If $x, y \in (0, 30)$ such that $\left[\frac{x}{3}\right] + \left[\frac{3x}{2}\right] + \left[\frac{y}{2}\right] + \left[\frac{3y}{4}\right] = \frac{11}{6}x + \frac{5}{4}y$ (where $[x]$ denotes greatest integer $\leq x$), then number of ordered pairs (x, y) is

(A) 0

(B) 2

(C) 4

(D) none of these

Q.6

Statement-1: The number of non-negative integral solution of
$x_1 + x_2 + \ldots + x_{20} = 100$
is $\begin{pmatrix} 120 \\ 20 \end{pmatrix}$.
Statement-2: The number of ways of distributing n identical objects among r persons giving zero or more objects to a person is $\begin{pmatrix} n+r-1 \\ r-1 \end{pmatrix}$.

(A) STATEMENT-1 isTrue, STATEMENT-2 is True; STATEMENT-2 is a correct explanation for STATEMENT-1

(B) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is NOT a correct explanation for STATEMENT-1

(C) STATEMENT-1 is True, STATEMENT-2 is False

(D) STATEMENT-1 is False, STATEMENT-2 is True

Q.7

The number of five digit numbers that can be formed by using digits $1, 2, 3$ only, such that three digits of the formed number are identical, is

(A) 30

(B) 60

(C) 120

(D) 90

Q.8

The total number of ways in which 5 balls of different colours can be distributed among three persons so that each person gets at least one ball is:

(A) 75

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(B) 150

(C) 210

(D) 243

Q.9

The number of ways we can put 5 different balls in 5 different boxes such that at most three boxes is empty, is equal to

(A) $5^5 - 5$

(B) $5^5 - 10$

(C) $2^5 - 2$

(D) none of these

Q.10

There are two urns. Urn A has 3 distinct red balls and urn B has 9 distinct blue balls. From each urn two balls are taken out and then transferred to the other. The number of ways in which this can be done is

(A) 66

(B) 108

(C) 3

(D) 36

Q.11

The number of ways of dividing 10 girls into two groups of 5 each so that two shortest girls are in the different groups is:

(A) 70

(B) 252

(C) 140

(D) 282

Q.12

The number of ways in which two teams A and B of 11 players each can be made up from 22 players so that two particular players are on the opposite sides is

(A) 369512

(B) 184755

(C) 184756

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(D) 369514

Q.13

Calculate the number of ways 5 balls can be placed in 3 boxes, such that no box remains empty, if balls as well as boxes are identical?

(A) 1

(B) 2

(C) 3

(D) 4

Q.14

Statement-1: If n is a natural number then $\frac{(n^2)!}{(n!)^{n+1}}$ is a natural number.
Statement-2: The number of ways of dividing mn students into m groups each containing n students is $\frac{(mn)!}{m!(n!)^m}$

(A) STATEMENT-1 isTrue, STATEMENT-2 is True; STATEMENT-2 is a correct explanation for STATEMENT-1

(B) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is NOT a correct explanation for STATEMENT-1

(C) STATEMENT-1 is True, STATEMENT-2 is False

(D) STATEMENT-1 is False, STATEMENT-2 is True

Q.15

The total number of ways in which 5 balls of different colours can be distributed among three persons so that each person gets at least one ball is:

(A) 75

(B) 150

(C) 210

(D) 243

Q.16

If 16 identical pencils are distributed among 4 children such that each gets at least 3 pencils.The number of ways of distributing the pencils is

(A) 15

(B) 25

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(C) 35

(D) 40

Q.17

The number of ways of distributing 8 identical balls in 3 distinct boxes so that none of the boxes is empty is

(A) ${}^{8}\mathrm{C}_{3}$

(B) 21

(C) 3^{8}

(D) 5

Q.18

The set $S=\{1,2,3,\ldots\ldots,12\}$ is to be partitioned into three sets A, B, C of equal size. Thus $A \cup B \cup C=S, A \cap B=B \cap C=A \cap C=\phi$.
The number of ways to partition S is

(A) $\frac{12!}{(4!)^3}$

(B) $\frac{12!}{(4!)^4}$

(C) $\frac{12!}{3!(4!)^3}$

(D) $\frac{12!}{3!(4!)^4}$

Q.19

The number of ways in which n distinct objects can be put into two different boxes so that no box remains empty, is

(A) $2^{n}-1$

(B) $n^{2}-1$

(C) $2^{n}-2$

(D) $n^{2}-2$

Q.20

The number of ways in which 8 distinct toys can be distributed among 5 children is

(A) 5^{8}

(B) 8^{5}

(C) ${}^{8}P_{5}$

(D) 40

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Q.1 Answer:

$\frac{1}{2}(p+1)(2k-p)$

Solution:

Let $xa's, yb's$ and $zc's$ be selected. Then number of selections is equal to the number of non-negative integral solutions of

$x+y+z=k$

If we take $x=l, 0 \leq x \leq p$ then $y+z=k-l$ and its number of solutions is $k-l+1$. Thus, the desired number of selections

$$\sum_{l=0}^{p}(k-l+1)=(k+1)(p+1)-\frac{1}{2}(p+1)(p+2)$$

$$=\frac{1}{2}(p+1)(2k-p).$$

Q.2 Answer:

${}^{27}C_3$

Solution:

Put $x=a-1, y=b-1, z=c-1, w=d-1$, then $a, b, c, d \geq 0$ and $(a-1)+(b-1)+(c-1)+(d-1)=20$

$\Rightarrow a+b+c+d=24$

The number of non-negative integral solutions of this equation is

${}^{24+4-1}C_{4-1}={}^{27}C_3$

Q.3 Answer:

1

Solution:

There is only one way in which five identical balls can be put in 10 boxes.

Q.4 Answer:

21

Solution:

If x_1 is the number of balls in the i th box, then we must have

$x_1+x_2+x_3=8$ **(1)**

where $x_i \geq 1$.

The number of solutions of (1) is ${}^{7}C_2=21$.

Q.5 Answer:

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none of these

Solution:

Let $\{x\} = x - [x]$ denote the fractional part of x. Note that $0 \le \{x\} < 1$. We can write the given equation as

$\frac{x}{3} - \left\{\frac{x}{3}\right\} + \frac{3x}{2} - \left\{\frac{3x}{2}\right\} + \frac{y}{2} - \left\{\frac{y}{2}\right\} + \frac{3y}{4} - \left\{\frac{3y}{4}\right\} = \frac{11}{6}x + \frac{5}{4}y$

$\Rightarrow \quad \left\{\frac{x}{3}\right\} + \left\{\frac{3x}{2}\right\} + \left\{\frac{y}{2}\right\} + \left\{\frac{3y}{4}\right\} = 0$

As each number on the L.H.S. lies in the interval $0 \le x < 1$, we must have

$\left\{\frac{x}{3}\right\} = \left\{\frac{3x}{2}\right\} = \left\{\frac{y}{2}\right\} = \left\{\frac{3y}{4}\right\} = 0$

$\Rightarrow \quad \frac{x}{3}, \frac{3x}{2}, \frac{y}{2}$ **and** $\frac{3y}{4}$ **must be integers.**

$\therefore x = 6, 12, 18, 24, \quad y = 4, 8, 12, 16, 20, 24, 28$

$\Rightarrow$ **Number of ordered pairs (x, y) equals $4 \times 7 = 28$.**

Q.6 Answer:

STATEMENT-1 is False, STATEMENT-2 is True

Solution:

The number of ways of distributing n identical objects among r persons giving zero of more objects to a person is equivalent to arranging n identical objects of one kind and $(r-1)$ identical objects of second kind in a row, which is equal to $\frac{(n+r-1)!}{n!(r-1)!} = \begin{pmatrix} n+r-1 \\ r-1 \end{pmatrix}$.

Next, the number of non-negative integral solutions of $x_1 + x_2 + \cdots + x_{20} = 100$ equals the number of ways of distributing 100 identical objects among 20 persons giving zero or more objects to a person, which equal $\begin{pmatrix} 100+20-1 \\ 20-1 \end{pmatrix} = \begin{pmatrix} 109 \\ 19 \end{pmatrix}$

Q.7 Answer:

120

Solution:

When exactly three digits are identical and the remaining two are different, then the number of such numbers

$\left({}^3C_1\right)\left(\frac{5!}{3!}\right) = 60$

When three digits are identical and the remaining two are also identical, then the number of ways

$= \left({}^3C_2\right)(2)\left(\frac{5!}{3!2!}\right) = 60$

Thus, the number of such numbers $= 120$.

Q.8 Answer:

150

Solution:

5 balls can be distributed to 3 persons by giving $(2, 2, 1)$ balls or by giving $(3, 1, 1)$ balls. Each of the above distribution has three such ways. Thus, the required number of ways

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$= (3)\left({}^5C_2\right)\left({}^3C_2\right)\left({}^1C_1\right) + 3\left({}^5C_3\right)\left({}^2C_1\right)\left({}^1C_1\right)$
$= (3)(10)(3)(1) + (3)(10)(2)(1) = 150$

Q.9 Answer:

$5^5 - 5$

Solution:

The number of ways of putting all the balls in exactly one $\text{box} = 5$

As for each ball there are 5 choices for the box, the number of ways of putting balls in the boxes $= 5^5$

Thus, the required number of ways of putting the balls in the boxes $= 5^5 - 5$

Q.10 Answer:

108

Solution:

The number of ways of transferring the balls
$= \left({}^3C_2\right)\left({}^9C_2\right) = (3)(36) = 108$

Q.11 Answer:

70

Solution:

Excluding two shortest girls, the remaining eight girls can be divided into two groups in $\frac{1}{2!}\left({}^8C_4\right)\left({}^4C_4\right) = 35$ **ways.**

The two shortest girls can be put in two different groups in 2 ways. Therefore, the required number of ways is 70.

Q.12 Answer:

369512

Solution:

We can choose 10 players from $(22-2)$ **players in** ${}^{20}C_{10}$ **ways and one player from 2 players in** ${}^2C_1$ **ways.**

$\therefore$ **required number of ways**
$= \frac{20!}{10!10!} \times 2 = 369512$

Q.13 Answer:

2

Solution:

3, 1, 1 & 2, 2, 1 → **two Methods only**

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Q.14 Answer:

STATEMENT-1 isTrue, STATEMENT-2 is True; STATEMENT-2 is a correct explanation for STATEMENT-1

Solution:

The number of ways of selecting students for the first group is $^{mn}C_n$**; for the second group is** $^{mn-n}C_n$ **and so on.**

$\therefore$ **the number of ways of dividing** (mn) **students into** m **numbered groups is**

$$\begin{aligned}&\left(^{mn}C_n\right)\left(^{mn-n}C_n\right)\cdots\left(^{n}C_n\right)\\=&\frac{(mn)!}{n!(mn-n)!}\frac{(mn-n)!}{n!(mn-2n)!}\cdots\frac{n!}{n!0!}\\=&\frac{(mn)!}{(n!)^m}\end{aligned}$$

As groups are not to be numbered, the desired number of ways is $\frac{(mn)!}{m!(n!)^m}$

$\therefore$ **Statement- 2 is true.**

For statement-1, put $m=n$**.**

Q.15 Answer:

150

Solution:

5 balls can be distributed to 3 persons by giving $(2,2,1)$ **balls or by giving** $(3,1,1)$ **balls. Each of the above distribution has three such ways. Thus, the required number of ways**

$$\begin{aligned}&=(3)\left(^5C_2\right)\left(^3C_2\right)\left(^1C_1\right)+3\left(^5C_3\right)\left(^2C_1\right)\left(^1C_1\right)\\&=(3)(10)(3)(1)+(3)(10)(2)(1)=150\end{aligned}$$

Q.16 Answer:

35

Solution:

Let a,b,c,d **be the number of things distubuted to each person such that** $a+b+c+d=16.$

we distibute 3 things to each person, then $3\times4=12$ **things are distributed. Thus only 4 things is left.**

We know that number of ways of distributing n **identical objects among** r **groups** $=\ ^{n+r-1}C_{n-1}$

Hence number of ways of ditubuting 4 identical things among 4 person $=\ ^{4+4-1}C_{4-1}=\ ^7C_3=\frac{7\times6\times5}{1\times2\times3}=35.$

Hence number of ways of ditributing 16 indentical things among 4 person such that each person gets atleast 3 things $=35$

Q.17 Answer:

21

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Solution:

Required number of ways
$=$ **coefficient of** x^8 **in** $\left(x+x^2+\ldots x^6\right)^3$
$\left[\because\right.$ **each box can receive minimum 1 and maximum 6 balls** $\left.\right]$
$=$ **coeff of** x^8 **in** $x^2\left(1+x+x^2+\ldots .+x^5\right)^3$
$=$ coeff **of** x^5 **in** $\left(\frac{1-x^6}{1-x}\right)^3$
$=$ **coeff of** x^5 **in** $(1-x)^{-3}$
$=$ **coeff of** x^5 **in** $\left(1+{}^3\mathrm{C}_1 x+{}^4\mathrm{C}_2 x^2+\ldots\right)$
$={}^7\mathrm{C}_5=21$

Q.18 Answer:

$\frac{12!}{(4!)^3}$

Solution:

Set $\mathrm{S}=\{1,2,3, \ldots \ldots 12\}$
$A \cup B \cup C=S, A \cap B=B \cap C=A \cap C=\phi$
$\therefore$ **The number of ways to partition**

$$
\begin{aligned}
& ={}^{12}C_4 \times {}^8C_4 \times {}^4C_4 \\
& =\frac{12!}{4!8!} \times \frac{8!}{4!4!} \times \frac{4!}{4!0!}=\frac{12!}{(4!)^3}
\end{aligned}
$$

Q.19 Answer:

2^n-2

Solution:

Each object can be put either in box B_1 **(say) or in box** B_2 **(say). So, there are two choices for each of the** n **objects. Therefore the number of choices for** n **distinct objects is** $2 \times 2 \times \ldots \times 2=2^n$**. Two of these choices correspond to either the first or the second box being empty. Thus, there are** 2^n-2 **ways in which neither box is empty.**

Q.20 Answer:

5^8

Solution:

Each toy can be distributed in 5 ways.
So, total number of ways $=5 \times 5 \times 5 \times 5 \times 5 \times 5 \times 5 \times 5=5^8$

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